Name - Niranjan Subhash More **PRN - 260840127018**

Cloud Services & Security
Assignment 1

Github

Q1 Create a Local git repository
Command = git init

Q2 Add Hello.java and Commit Hello.java

```
pgcp-hpcsa@vedant:~/cloud-services$ nano Hello.java
pgcp-hpcsa@vedant:~/cloud-services$ git add Hello.java
pgcp-hpcsa@vedant:~/cloud-services$ git commit -m "Init Commit" Hello.java
[main a8c63a0] Init Commit
 1 file changed, 5 insertions(+)
 create mode 100644 Hello.java
```

Q3 Add file1.txt,Modify Hello.java and Commit Both files

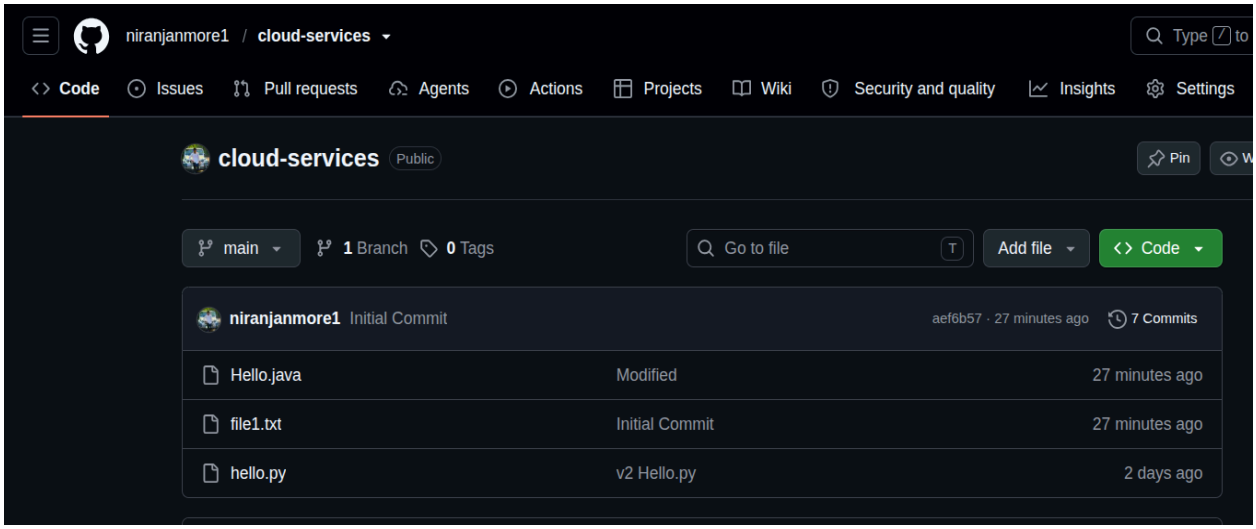
```
pgcp-hpcsa@vedant:~/cloud-services$ nano file1.txt
pgcp-hpcsa@vedant:~/cloud-services$ git add file1.txt
pgcp-hpcsa@vedant:~/cloud-services$ nano Hello.java
pgcp-hpcsa@vedant:~/cloud-services$ git commit -m "Modified" Hello.java
[main 676ff7e] Modified
 1 file changed, 3 insertions(+), 5 deletions(-)
pgcp-hpcsa@vedant:~/cloud-services$ git commit -m "Initial Commit" file1.txt
[main aef6b57] Initial Commit
 1 file changed, 1 insertion(+)
 create mode 100644 file1.txt
```

Q4 Create Github Repository Push all Files From Local Repository to Github

Steps 1 - Go to https//:github.com
Step 2 - Login With Username and Password
Step 3 - Create New Repository

```
pgcp-hpcsa@vedant:~/cloud-services$ git remote add origin https://github.com/niranjanmore1/cloud-services.git
error: remote origin already exists.
pgcp-hpcsa@vedant:~/cloud-services$ git push -u origin --all
Username for 'https://github.com': niranjanmore1
Password for 'https://niranjanmore1@github.com':
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 12 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (9/9), 1.01 KiB | 259.00 KiB/s, done.
Total 9 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/niranjanmore1/cloud-services.git
   c832fb6..aef6b57  main -> main
Branch 'main' set up to track remote branch 'main' from 'origin'.
```

Verify On Github -



Docker

Q1 Create a Docker image with nginx as base image The Image Should Copy index.html within nginx Container.Run The Container & Verify if Your index.html Was Created in nginx container.

```
pgcp-hpcsa@vedant:~/docker-q1$ cat > Dockerfile <<'EOF'
FROM docker.io/library/nginx:latest
COPY index.html /usr/share/nginx/html/index.html
EOF
```

```
pgcp-hpcsa@vedant:~/docker-q1$ docker build -t my-nginx .
Emulate Docker CLI using podman. Create /etc/containers/nodocker to quiet msg.
STEP 1/2: FROM docker.io/library/nginx:latest
Trying to pull docker.io/library/nginx:latest...
Getting image source signatures
Copying blob ecc510c1e359 skipped: already exists
Copying blob 792a76c59dbd done
Copying blob c9ec797c69ed done
Copying blob 6a6757ab48b1 done
Copying blob b19352c60515 done
Copying blob 188e9077f350 done
Copying blob 113d9139ada2 done
Copying config ca5e66f340 done
Writing manifest to image destination
Storing signatures
STEP 2/2: COPY index.html /usr/share/nginx/html/index.html
COMMIT my-nginx
--> f9450fd1c3a
Successfully tagged localhost/my-nginx:latest
f9450fd1c3a6852bef707fa526cb6520631e0cf7a67413d219d61bbdb2e46c86
```

```
pgcp-hpcsa@vedant:~/docker-q1$ docker rm -f my-nginx-container
Emulate Docker CLI using podman. Create /etc/containers/nodocker to quiet msg.
e43a6ae8b3a22f8e73faab11a7dbeeab72911af4007f55a4117fabd547a54b53
pgcp-hpcsa@vedant:~/docker-q1$ docker run -d --name my-nginx-container -p 8081:80 my-nginx
Emulate Docker CLI using podman. Create /etc/containers/nodocker to quiet msg.
744732ee7c85aa5538ddd0bfb692e73cbdc023c9c15d270adc588b5d44296548
pgcp-hpcsa@vedant:~/docker-q1$ sleep 2
pgcp-hpcsa@vedant:~/docker-q1$ docker ps
Emulate Docker CLI using podman. Create /etc/containers/nodocker to quiet msg.
CONTAINER ID  IMAGE                     COMMAND               CREATED         STATUS             PORTS                 NAMES
744732ee7c85  localhost/my-nginx:latest  nginx -g daemon o...  43 seconds ago  Up 43 seconds ago  0.0.0.0:8081->80/tcp  my-nginx-container
```

```
pgcp-hpcsa@vedant:~/docker-q1$ docker exec my-nginx-container ls -l /usr/share/nginx/html/
Emulate Docker CLI using podman. Create /etc/containers/nodocker to quiet msg.
total 8
-rw-r--r-- 1 root root 497 Oct  1 17:19 50x.html
-rw-rw-r-- 1 root root 129 Oct 16 03:39 index.html
pgcp-hpcsa@vedant:~/docker-q1$ docker exec my-nginx-container cat /usr/share/nginx/html/index.html
Emulate Docker CLI using podman. Create /etc/containers/nodocker to quiet msg.
<!DOCTYPE html>
<html>
<head><title>Docker nginx</title></head>
<body>
  <h1>Hello from my nginx container!</h1>
</body>
</html>
```

```
pgcp-hpcsa@vedant:~/docker-q1$ curl http://localhost:8081
<!DOCTYPE html>
<html>
<head><title>Docker nginx</title></head>
<body>
  <h1>Hello from my nginx container!</h1>
</body>
</html>
```



Hello from my nginx container!

Q2 Create a Dockerfile to install curl on alpine build and run the container.The curl command should access google.com on running the container.

```
pgcp-hpcsa@vedant:~/docker-q1$ mkdir -p ~/docker-q2 && cd ~/docker-q2
pgcp-hpcsa@vedant:~/docker-q2$ cat > Dockerfile <<'EOF'
FROM docker.io/library/alpine:latest
RUN apk add --no-cache curl
CMD ["curl", "-sL", "-o", "/dev/null", "-w", "HTTP status: %{http_code}\n", "http://google.com"]
EOF
pgcp-hpcsa@vedant:~/docker-q2$ cat Dockerfile
FROM docker.io/library/alpine:latest
RUN apk add --no-cache curl
CMD ["curl", "-sL", "-o", "/dev/null", "-w", "HTTP status: %{http_code}\n", "http://google.com"]
```

```
pgcp-hpcsa@vedant:~/docker-q2$ docker build -t alpine-curl .
Emulate Docker CLI using podman. Create /etc/containers/nodocker to quiet msg.
STEP 1/3: FROM docker.io/library/alpine:latest
Trying to pull docker.io/library/alpine:latest...
Getting image source signatures
Copying blob e2de96513ba9 skipped: already exists
Copying config 320994c3b9 done
Writing manifest to image destination
Storing signatures
STEP 2/3: RUN apk add --no-cache curl
(1/9) Installing brotli-libs (1.2.0-r1)
(2/9) Installing c-ares (1.34.8-r0)
(3/9) Installing libunistring (1.4.2-r0)
(4/9) Installing libidn2 (2.3.8-r0)
(5/9) Installing nghttp2-libs (1.70.0-r0)
(6/9) Installing libpsl (0.21.5-r3)
(7/9) Installing zstd-libs (1.5.7-r2)
(8/9) Installing libcurl (8.22.0-r0)
(9/9) Installing curl (8.22.0-r0)
Executing busybox-1.37.0-r31.trigger
OK: 13.1 MiB in 25 packages
--> 81ef430448f
STEP 3/3: CMD ["curl", "-sL", "-o", "/dev/null", "-w", "HTTP status: %{http_code}\n", "http://google.com"]
COMMIT alpine-curl
--> 78b16b28e96
Successfully tagged localhost/alpine-curl:latest
78b16b28e9653dd852874bb2cb56eff326c95e8235957415c8cdb8b1b661cf6b
```

```
pgcp-hpcsa@vedant:~/docker-q2$ docker images | grep alpine-curl
Emulate Docker CLI using podman. Create /etc/containers/nodocker to quiet msg.
localhost/alpine-curl                   latest           78b16b28e965  28 seconds ago  14.1 MB
```

```
pgcp-hpcsa@vedant:~/docker-q2$ docker run --rm alpine-curl
Emulate Docker CLI using podman. Create /etc/containers/nodocker to quiet msg.
HTTP status: 200
```

```
pgcp-hpcsa@vedant:~/docker-q2$ docker run --rm alpine-curl curl -sIL http://google.com
Emulate Docker CLI using podman. Create /etc/containers/nodocker to quiet msg.
HTTP/1.1 301 Moved Permanently
Location: http://www.google.com/
Content-Type: text/html; charset=UTF-8
Content-Security-Policy-Report-Only: object-src 'none';base-uri 'self';script-src 'nonce-DzMEk7oy06dS-lGJ7-_G8w' 'strict-dynamic' 'report-sample' 'unsafe-eval' 'unsafe-inli
ne' https: http:;report-uri https://csp.withgoogle.com/csp/gws/other-hp
Date: Fri, 16 Oct 2026 04:12:06 GMT
Expires: Sun, 15 Nov 2026 04:12:06 GMT
Cache-Control: public, max-age=2592000
Server: gws
Content-Length: 219
X-XSS-Protection: 0
X-Frame-Options: SAMEORIGIN

HTTP/1.1 200 OK
Content-Type: text/html; charset=ISO-8859-1
Content-Security-Policy-Report-Only: object-src 'none';base-uri 'self';script-src 'nonce-9iFCxmO0_8CZJOGbDDALjg' 'strict-dynamic' 'report-sample' 'unsafe-eval' 'unsafe-inli
ne' https: http:;report-uri https://csp.withgoogle.com/csp/gws/other-hp
P3P: CP="This is not a P3P policy! See g.co/p3phelp for more info."
Date: Fri, 16 Oct 2026 04:12:06 GMT
Server: gws
X-XSS-Protection: 0
X-Frame-Options: SAMEORIGIN
Expires: Fri, 16 Oct 2026 04:12:06 GMT
Cache-Control: private
Set-Cookie: __Secure-STRP=ABNhA6iCTBQvdaHmb73ADpQydG2cUZX9v_BpGd7OK4s9HN9t1bo35M19NmDUT3FeqBmdd9BjWL3HyiufiMH2anABnCQh9vsdMmPj; expires=Fri, 16-Oct-2026 04:17:06 GMT; path=
/; domain=.google.com; Secure; SameSite=strict
Set-Cookie: AEC=Aaa9EJpyQBjy5XJwYW5ea3LVp0nzT7Q40umoO0Qhd3bAtCodRg1Dy79JuQ; expires=Wed, 14-Apr-2027 04:12:06 GMT; path=/; domain=.google.com; Secure; HttpOnly; SameSite=la
x
Set-Cookie: NID=Cu8BCAESoQEBOxGDSFIih7q3witO0o7YztveXKWvKnjxyC_iFiknycuAkpgBSjL6nILAcj6_JrI7J2mjhnr43hJPtS2spuhdftds6uASU-XTupj6FsPMFPUBSg_PvdvwBrPbWRXB7XskrCuJvmYlYenWjUZ3
liulaEz4sdjUhNwJEfoZl5vAb1HYzG8WIDKKlmtYNPIQL5iILMFG6bITx7e2e3Yl1GlooygBMkUB0qwHz0736wydm4p3PJLMh5W1_JcvUMY8n_yATh2VsE44KueGAEIYY11kV54OcgbASGAaLHSfs6fuhCbpqBjgt1zPPw0; exp
ires=Sat, 17-Apr-2027 04:12:06 GMT; path=/; domain=.google.com; HttpOnly
Transfer-Encoding: chunked
```